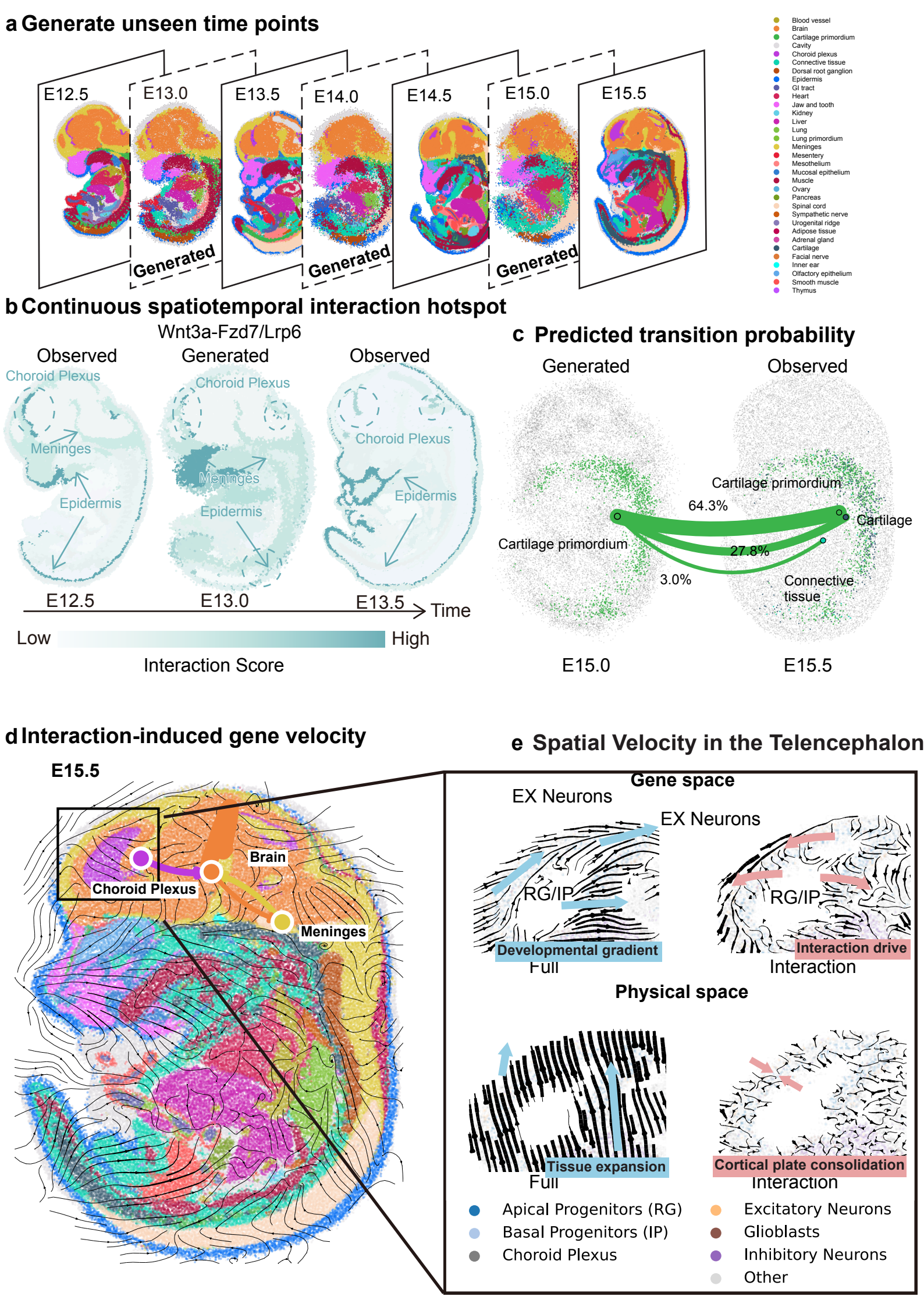

a Generate unseen time points
E12.5
E13.0
E13.5
E14.0
E14.5
E15.0
E15.5
Generated
Generated
Generated
Blood vessel
Brain
Cartilage primordium
Cavity
Choroid plexus
Connective tissue
Dorsal root ganglion
Epidermis
GI tract
Heart
Jaw and tooth
Kidney
Liver
Lung
Lung primordium
Meninges
Mesentery
Mesothelium
Mucosal epithelium
Muscle
Ovary
Pancreas
Spinal cord
Sympathetic nerve
Urogenital ridge
Adipose tissue
Adrenal gland
Cartilage
Facial nerve
Inner ear
Olfactory epithelium
Smooth muscle
Thymus
b Continuous spatiotemporal interaction hotspot
Wnt3a-Fzd7/Lrp6
Observed
Generated
Observed
Choroid Plexus
Meninges
Epidermis
E12.5
E13.0
E13.5
Time
Low
High
Interaction Score
c Predicted transition probability
Generated
Observed
Cartilage primordium
64.3%
Cartilage
Cartilage primordium
27.8%
3.0%
Connective tissue
E15.0
E15.5
d Interaction-induced gene velocity
E15.5
Brain
Choroid Plexus
Meninges
e Spatial Velocity in the Telencephalon
Gene space
EX Neurons
EX Neurons
RG/IP
RG/IP
Developmental gradient
Interaction drive
Full
Interaction
Physical space
Tissue expansion
Cortical plate consolidation
Full
Interaction
Apical Progenitors (RG)
Basal Progenitors (IP)
Choroid Plexus
Excitatory Neurons
Glioblasts
Inhibitory Neurons
Other